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Fat Distinguish and Visceral Fat Expansion

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Abstract

Compared with traditional medical imaging method such as CT and MRI, digital Human Phantom (DHP) is a useful and convenient method to simulate the situation of a human body. Although some DHPs are generated previously, DHP for a overweight person's torso is not available nowadays since some certain features about human's torso.

The increasing of fat is the most simple way to simulate a overweight person. According to previous students work, a DHP of subcutaneous fat increasing is generated and a rough method to increase the volume of visceral fat is put forward.

In this project, fat distinguish algorithm and fat expansion algorithm are developed. The first algorithm aims to determine the type of fat tissue precisely and the second algorithm aims to apply fat expansion on visceral fat. Some sub-algorithms are developed in order to improve the performance of the initial algorithms.

This report will introduce those two algorithms and show the outcomes of those algorithms. In the end of this report, the future improvements and future work are mentioned.

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1 Introduction

1.1 Background Information

Although in ¹the early era, extra weight are considered as healthy and wealthy, obesity are finally be officially regarded as a kind of disease in the last 20 century [1, 2]. One criterion to distinguish an obesity patient and a normal person is to calculate his or her Body Mass Index (BMI) [1]. Different body shape, such as health, overweight and obesity, can be separated according to BMI.

Visceral fat increasing is a crucial characteristic for a overweight person or a obesity patient compared with a health normal person. Too much visceral fat can cause a lot of disease like vascular or cardiac complications of hypertension [3], arterial stiffness [4], glucose intolerance [5] and so on. By monitoring the visceral fat and other health criteria in a early date, some early disease can be found and some relative operations can be taken to save lives and save expenses.

Centuries ago, people do not know how to monitor the conditions inside the body in a non-anatomy way before the X-Ray shows its ability on medical imaging. ⁶Computer Tomography (CT), Magnetic Resonance Image (MRI) and Positron Emission Tomography (PET) are three main methods ⁵to generate ⁴the medical images. CT mainly uses X-Ray. The radiation is injected from one side of the body and they are detected by the electrophoton detector on the other side. The received signals are transformed into images by mathematical calculations. For PET, patients are injected with a radioactive preparation labeled with certain positrons. The results are presented by bio-molecular metabolism. Compared with CT and PET, MRI shows a great ability on soft tissue (such as fat tissue) testing. ²MRI is based on the principle that atomic nuclei with magnetic moments can produce transitions between energy levels under the action of a magnetic field. Although MRI is more expensive than CT, its harmless characteristic and its ability to detect soft tissue let it become a very common method in medical imaging.

Since PET is a very new and very expensive technology, CT and MRI are always considered as the first option when a doctor wants to create a set of image for a patient. Typically, both methods should be taken in order to get more information for doctors. However, taking CT and MRI together is time-consuming and sometimes the expenses for CT and MRI is not affordable for some patients. Moreover, for a patient who needs to take the scans frequently, it is also unbearable. So Digital Human Phantom (DHP) is developed as an alternative approach. DHP saves

the biophysical information for some certain parts of the human body digitally, so it could be operated and analyzed by software. DHP has a good portability for various conditions. So it is widely used in medical field. From previous peoples work, the DHP for heart [6], lungs [7] and brains [8] are already built and used in real medical circumstances. The DHP for obesity patients is hard to build because of the uncertainty of organs and individual differences caused from races, age, gender and some other feature. But, still, the basic demo should be created because it can simulate the approximate model for a obesity patient, which is also helpful for prediction and simulation.

Compared with traditional medical imaging method, DHP is the cheapest and the most flexible approach. Only the original data should be collected. In order to solve the problem mentioned before, many DHPs should be created, each of them present a type of people. Once the patient fits the features of one DHP, this DHP can be used to simulate and predict this patient's health conditions.

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1.2 Aims and objectives

This project aims to develop algorithms to build a visceral fat expansion DHP for a man's torso. MATLAB is mainly used to operate the data and Python is used to support MATLAB operation. Two separate algorithms are used in this project. One aims to distinguish the visceral fat and the subcutaneous fat, the other aims to increase the visceral fat based on the results of the first algorithm. The first algorithm can also be used in other researches directly. The original data used for generating the DHP in this project is collected from a 55 years old, healthy, normal body shape Japanese man.

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The following objectives should be finished in this project:

- 1) Understand the Portable Gray Map (PGM) format for storing medical image data
- 2) Change PGM format data into a readable format which could be shown and manipulated by MATLAB or Python
- 3) Research the basic anatomy for a human body, and search for the necessary data for later operations
- 4) Apply an algorithm to distinguish two kinds of fat and make it suitable for every slices of data

- 5) Research the effect of visceral fat expansion, and its expressive form applied on image
- 6) Evaluate the outcomes in different parameters used in the algorithm
- 7) Indicate the direction of future work

1.3 Deliverables

- 1) Dissertation Report: The report for this project illustrates everything related to this project, which include the related literature research, the methods to implement the algorithm, outcomes of the algorithms and future works.
- 2) Program: The programs for this project are mainly running on MATLAB R2021b environment. Also one Python program is used for generate a more easy-to-use format of data, which runs under Python 3.8.1 environment. All the programs are based on Windows 10 OS.
- 3) GUI: The outcomes are shown through a GUI, which allows user to modify target DHP by setting some specific parameters.

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2 Literature Review

2.1 Digital Human Phantoms (DHPs)

Digital Human Phantoms can be regarded as a set of models that present the data from different sections of human's body. There are two main methods to build DHP, one is by using anatomy facts and mathematical modelling, the other is by using the medical images obtained from CT or MRI in clinic experiments [9]. Because of the different features, those two methods are always used together in the practical circumstances. Once the models are generated, some certain body situation can be simulated by manipulating those DHP models. DHP models have a good portability and robustness, so they are widely used in the medical field. Now, DHP models are not only used in medical field, but also used in many other area. For example, molecular imaging [10] and non-rigid motion [11] can also be researched by using DHP models.

There are a lot of DHP models available nowadays, which will suit different patient situation. Because it is hard to do dissection on brain, brain is one of the earliest digital phantom generated in order to simulate the approximate situation different brain diseases [12]. DHP for other certain organs like heart [13] and lung [14] are also built after brain model. Although there are some models for chest part [15] and the abdomen part [16] of the body, but most of them are using a 4-D approach [15], which means they used more data and more biophysical research on the motion of tissues. However, in this project, only basic (non-4-D) DHP will be concerned since data limited and time limited.

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The data used in this project was provided by RIKEN (Saitama, Japan) under a no-disclosure agreement between RIKEN and the University of Manchester. The usage was approved by RIKEN ethical committee [17].

2.2 Portable Gray Map (PGM) Format

The first thing to do in this project is to understand the format for storing the data. There are a lot of formats in medical field such as DICOM (.dcm) and NIFTI (.nifti). But in this project, a more simple format called portable gray map is used for storing the data. A normal .pgm file has five components. The illustration is shown as followed.

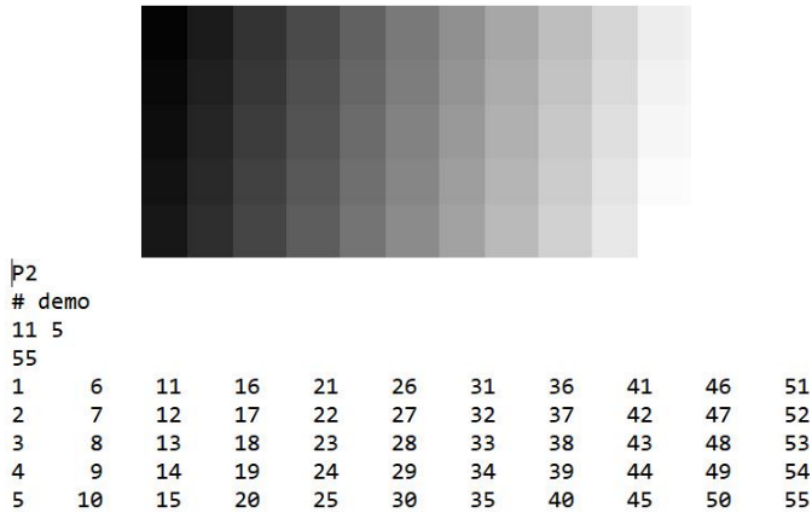


Figure 2.1 Illustration of PGM File

As the name says, the .pgm file shows a gray scale image. The first line of .pgm file expresses two types of image, 'P2' means gray scale image, while 'P5' means binary image. The data used in this project are all in 'P2' type. The second line is the comment line. It could be ignored during processing. The number pair in the third line means the size of this image in pixels. The number in the fourth line defines the maximum number, which represents white, in this file. After the program reads this number, it knows the number range to present data in this image is from zero to this number and then transform data to other data type such as uint8. The last section is the data for presenting the image.

2.3 Data Used in This Project

There are 500 slices of .pgm files in this project to present the torso part and of course they are in the same size which is 265*490 pixels. The maximum value is 80. The value part of those data has some difference compared with the illustration in Figure 2.1. The data are in a column which contains 245000 values instead of a matrix shape. When dealing those arrays, the certain program will reshape them in to a 265*490 matrix.

There are different values which represent various tissues. More details could be found through the anatomy research.

The resolutions in this data set should also be noted. The resolution in one slice is 1mm/pixel and the resolution between slices is 1mm/slice. By combining those resolutions with some medical research, this project can be implemented in a reasonable manner.

2.4 Anatomy Research

Since this project is related with medical background, some medical research must be done. According to *Atlas of Human Anatomy*, visceral fat most store in the fat cell and surround by some membranes [18]. Those membranes are also regarded as fat in the data used in this project.

The relationship between bones and muscles is also important. No matter how much visceral fat a person has, bones must connected with muscles. For a person whose the weight changing in a range (not too much changing), the bones are stable compared with muscles. This fact can also prove that visceral fat all distributed in membranes.

More anatomy researches are presented in the latter chapter.

3 Fat Distinguish Algorithm (FDA)

3.1 Pre-processed

3.1.1 Get All Data

The original data provided in this project is collected from a healthy normal Japanese man aged around 55. They are stored into 500 slices of .pgm files. As it says in Chapter 2, .pgm files has a typical type to store data, which will make them not easy to visualize. The first step is removing all the prefix and reshape them into a 490*290 pixels shape. Since there are 500 slices in total, stack every slice into one file, then this 245000*265 pixels' file concludes all the data.

3.1.2 Reassign the Gray Values

Original data set different organ in different gray value. But in this project, fat is the most important thing that should be concerned. Also two gray values should be reserved for assigning the visceral fat and subcutaneous fat. The newly assigned values are shown in Table 3.1 and reassigned image is shown in Figure 3.1.

Table 3.1 New Values for Different Tissues

Air	Skin	Muscle	Organ	Bone	Fat	Vis-Fat	Sub-Fat
0	20	30	40	50	70	10	80

Note that the maximum value is 80.

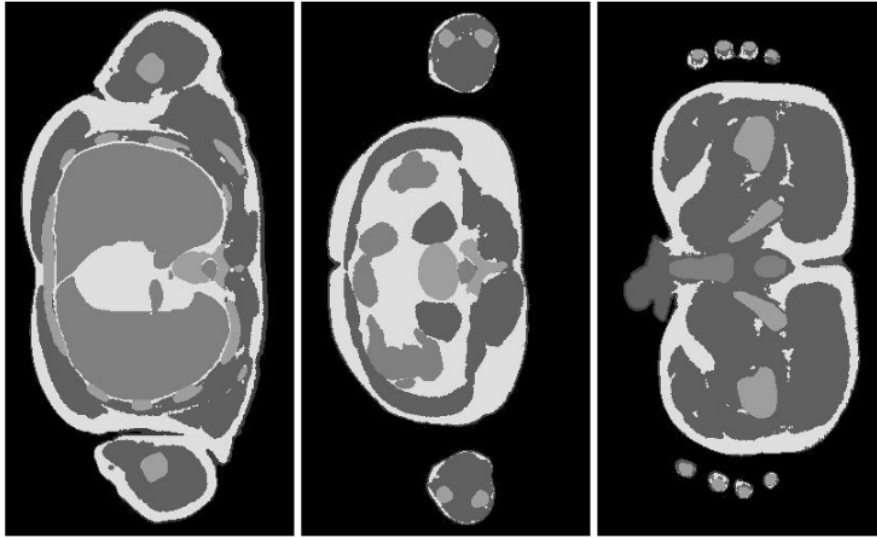


Figure 3.1 Images after Reassign the Values of Tissues (Slice 1, 300, 500)

3.2 Initial Fat Distinguish Algorithm (FDA)

3.2.1 Disk Filter and Dilation Operation

In this project, visceral fat is expected ³to expand in a circular manner, so a circular filter is concerned. Euclidean distance is the most precise way, but it needs more calculation. Since the radius of the filter used in this project should not be very large (this will be mentioned very soon), a octagonal disk filter can be used to approximate the circular filter. The way to generate this disk filter is to calculate the sum of the horizontal distance and the vertical distance between current pixel and central pixel. If this sum is bigger than radius, the value of this pixel is set to zero and vise versa. Note that when the radius is small, disk filter can be regard as circular filter. Demonstration of disk filter used in this project is shown as Figure 3.2.

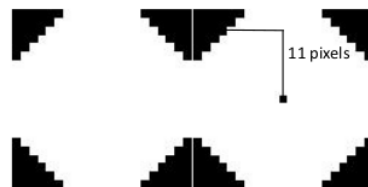


Figure 3.2 Illustration of the Octagonal Filter (with radius 11 pixels)

Dilation is a morphological method in image processing. Simply, it will expand the 'white' part following a certain structure element. By using disk filter as structure element, a circular expansion can be obtained. Demonstration of dilation method is shown as Figure 3.3

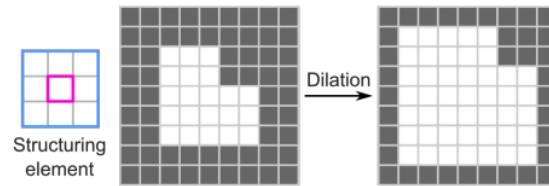


Figure 3.3 Illustration of Dilation Operation [20]

3.2.2 Initially Distinguish Subcutaneous Fat and Visceral Fat

In order to distinguish two kinds of fat automatically, a stable character should be used. Because the data is collect from a normal healthy man, the mean thickness of subcutaneous adipose tissue is around 10mm[19]. Since the resolution of the data is 1mm/pixel, 10 pixels are chosen as the width of subcutaneous fat.

To distinguish two kind of fat, dilation can be used. The method is shown in Chart 3.1.

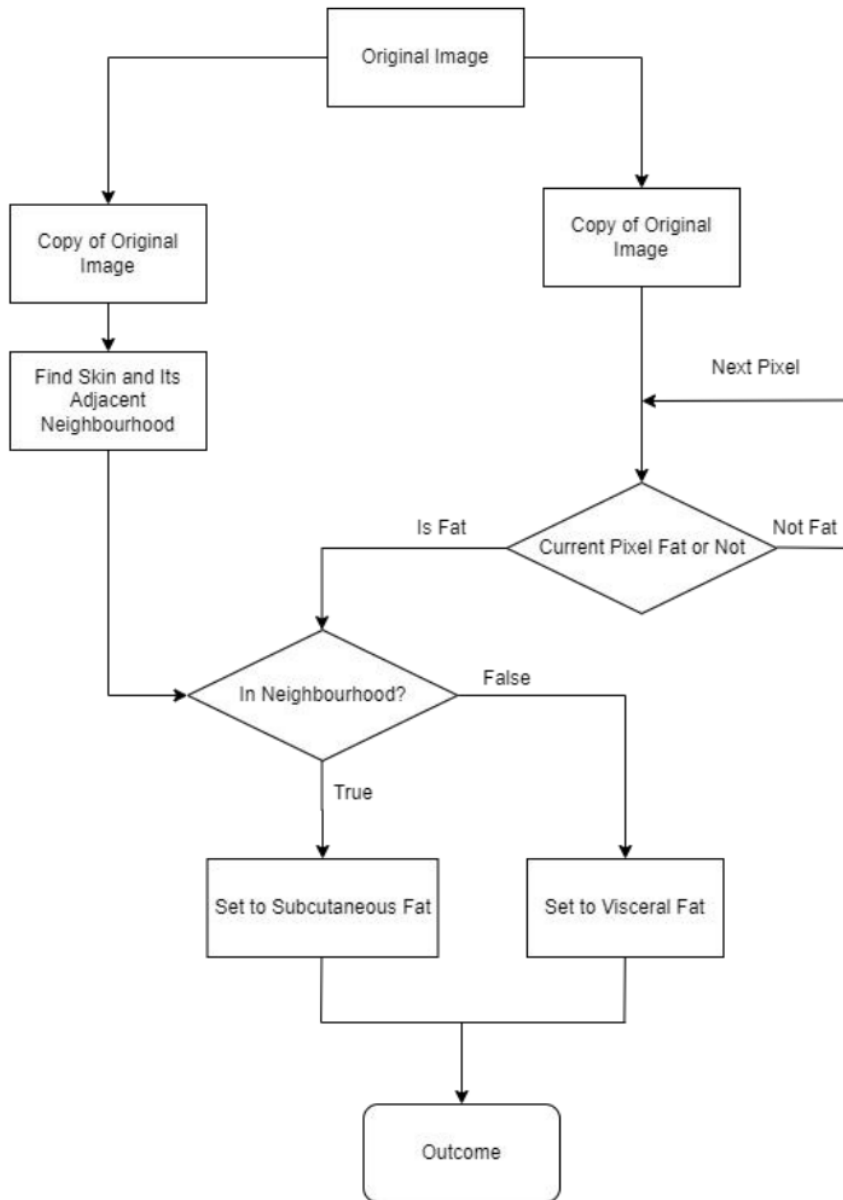


Chart 3.1

As Figure 3.4 shows, creating a new binary image (Figure 3.4-a) where only skin pixels are set to 1, else pixels are set to 0. By applying dilation method on this binary image, the possible range of subcutaneous fat is obtained (Figure 3.4-b). Next, make a judgement: if fat in this range, it should be subcutaneous adipose tissue, else is regarded as visceral fat.

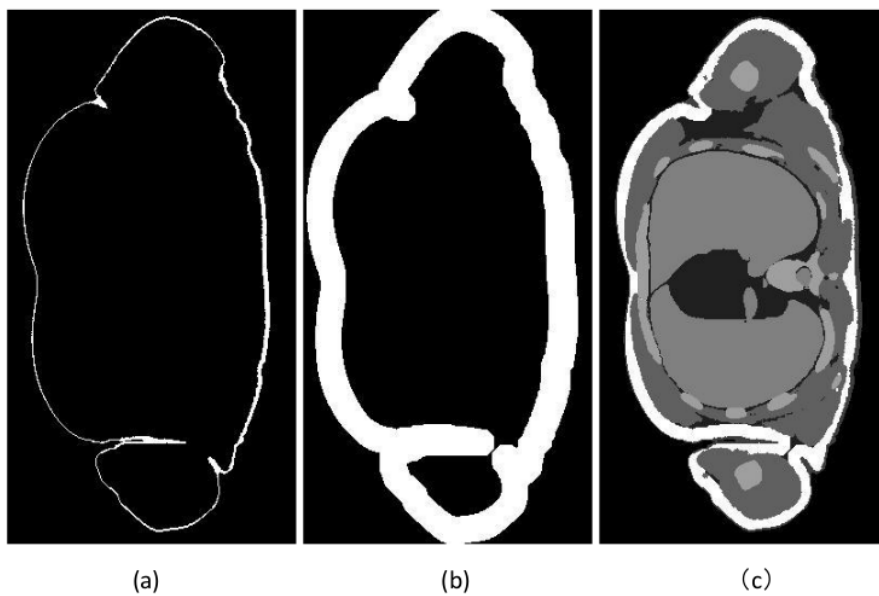


Figure 3.4 The Intermediate Results and Final Result during Initial FDA (Slice 1)

3.3 Label Method to Improve FDA

From an analysis of result image Figure 3.4-c, a large portion of subcutaneous fat are not distinguished, as it shown in Figure 3.5. So this sub-algorithm is developed to deal with this problem.

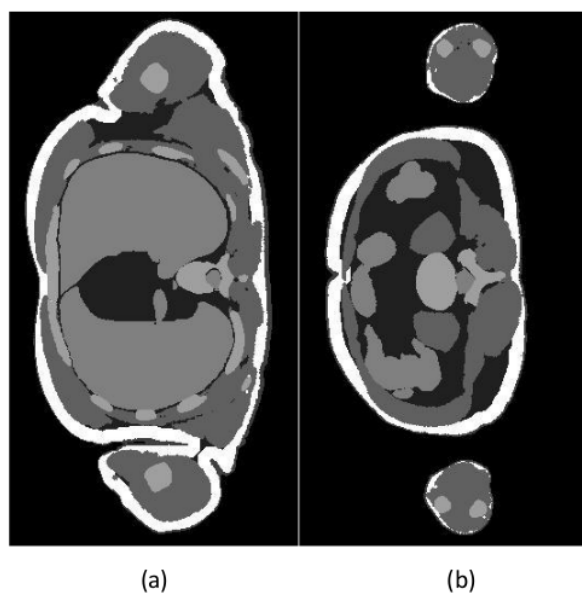


Figure 3.5 Imperfections of Initial FDA

3.3.1 Introduction of Label Matrix

Label matrix is used in label method to improve FDA. ⁴ 'A label matrix labels objects or connected components in a binary image with unique integer values [21].' In another word, label matrix assign the connected pixels with the same value.

3.3.2 Initial Label Method and Improved Label Method

It is easy to notice that in Figure 3.5a, subcutaneous adipose tissue are connected, so they should be assign with the same label. By set values of pixels which have same label number in label matrix, connected fat tissues are assigned with the same categories. The method shows in Chart 3.2.

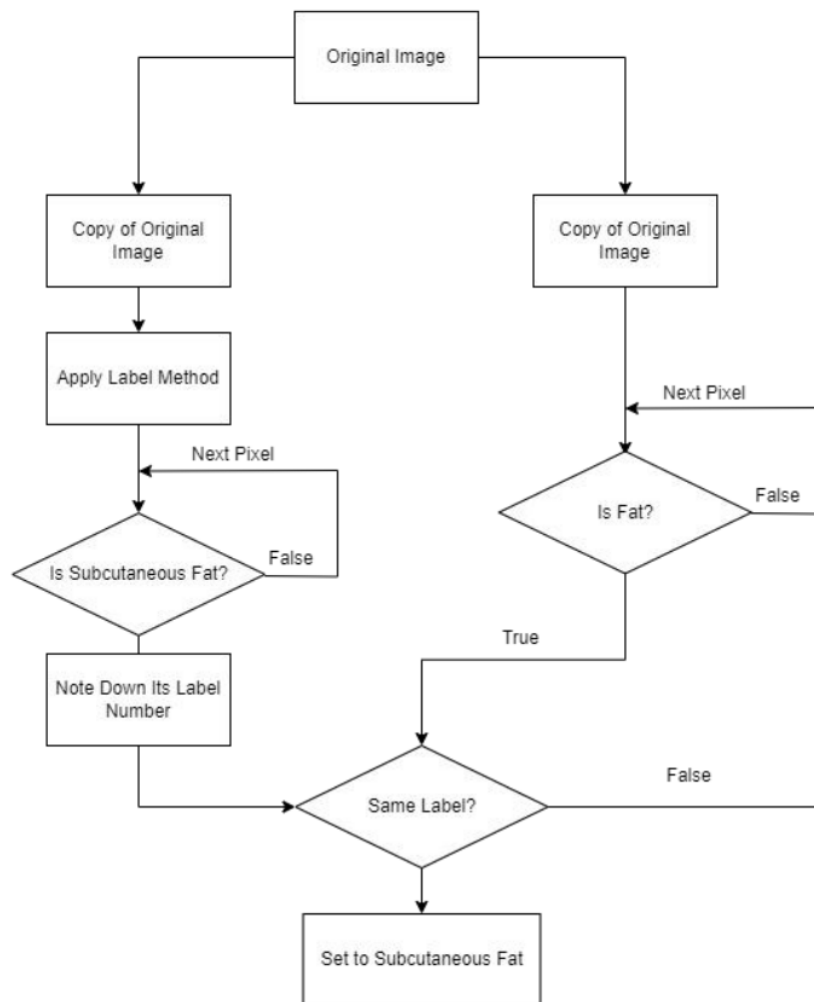


Chart 3.2

The results of initial label method applied in both chest part and abdomen part is shown in ⁹in Figure 3.6.

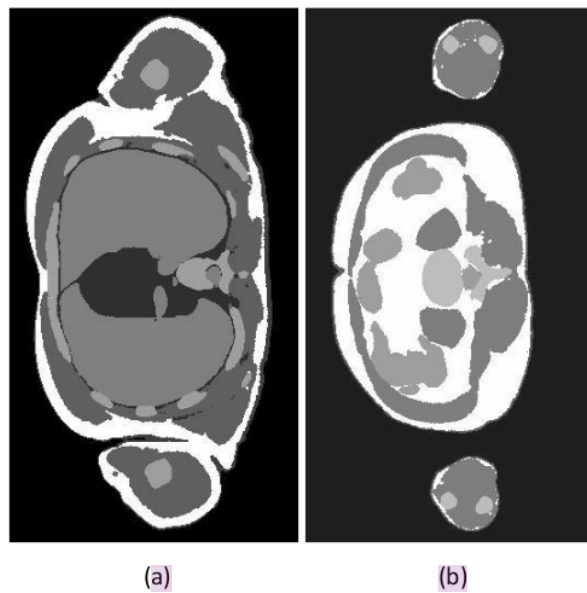


Figure 3.6 Performance of Label Method Applied on Slice 1 and Slice 300

Because of surrounded by muscles, the result in chest part is quite good. But there are no continuous muscles in abdomen part, which means it will cause unusual output by using label method. So a improved label method is shown in Chart 3.3

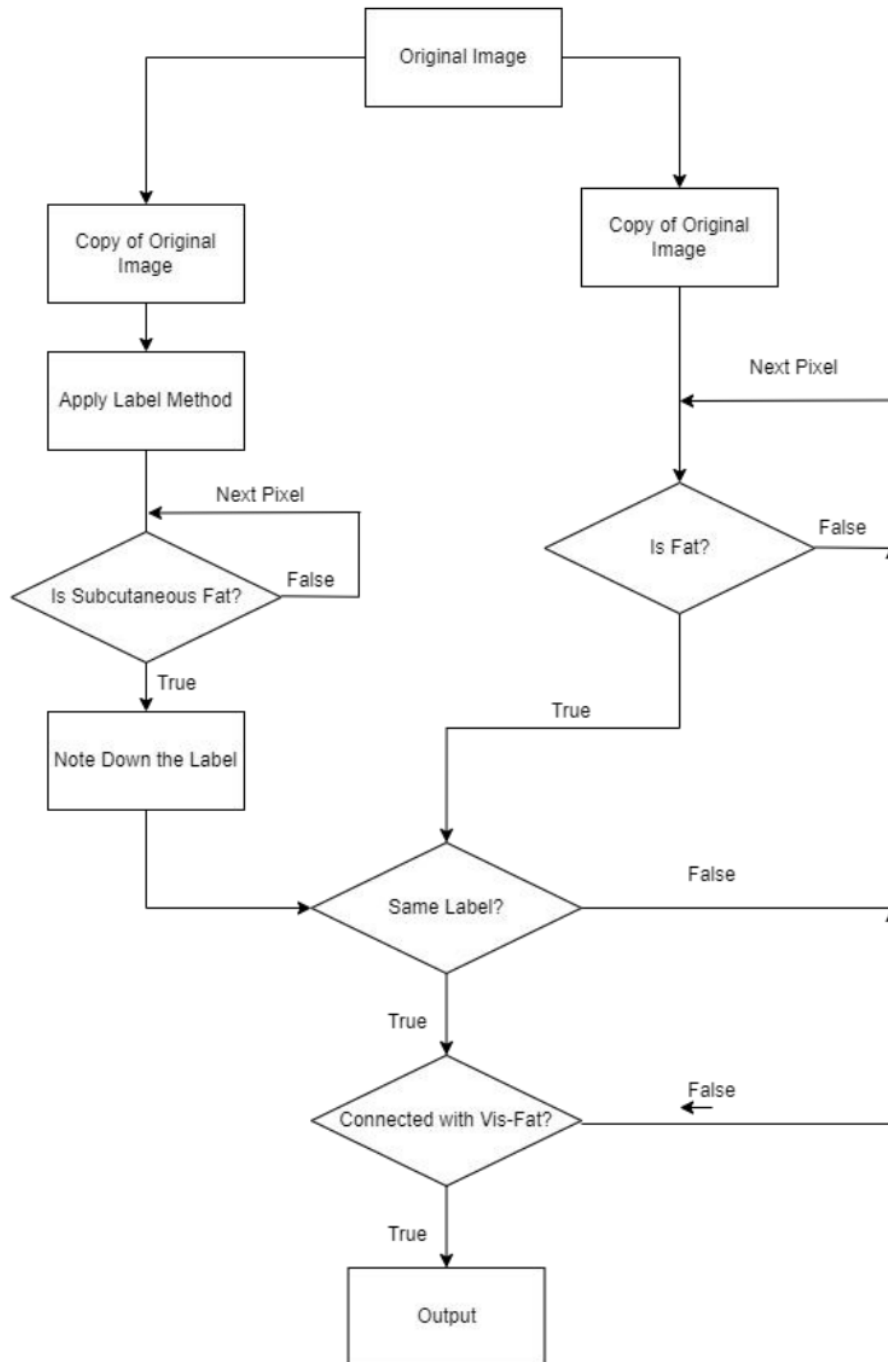


Chart 3.3

This improved method needs one more judgement to figure out whether current label connected with 'real' visceral fat or not. Firstly, use dilation method to find out the 'real' visceral

fat. Then, if current label connected with 'real' visceral fat, this algorithm will change nothing. It means for most abdomen part, the output is the same as the initial FDA.

3.4 Polygonal Method to Improve FDA

The results discussed in last section shows that the label method make no improvement at the abdomen part. A abdomen specific method should be introduced.

3.4.1 The Convex Hull Method

Noticed that the transversal plane of abdomen is always in a convex form, all the fat tissue inside this convex hull should be distinguished as visceral fat. The problem now convert into how to find this convex hull.

The easiest way is to find every muscle pixels, then find the convex hull that concludes all those pixels. Outside of this hull are set to subcutaneous fat and the else are set to visceral fat.

This method will improve FDA, but still not good enough because during choosing muscle pixels, the muscle connected with subcutaneous fat is counted, which are harmful for building the convex hull.

3.4.2 Polygonal Method and Its Improvement

In order to solve the problem in the convex hull method, the inside wall of muscle should be found. In another word, only the muscle pixels that are adjacent to visceral fat should be recorded.

The method is shown in Chart 3.4.

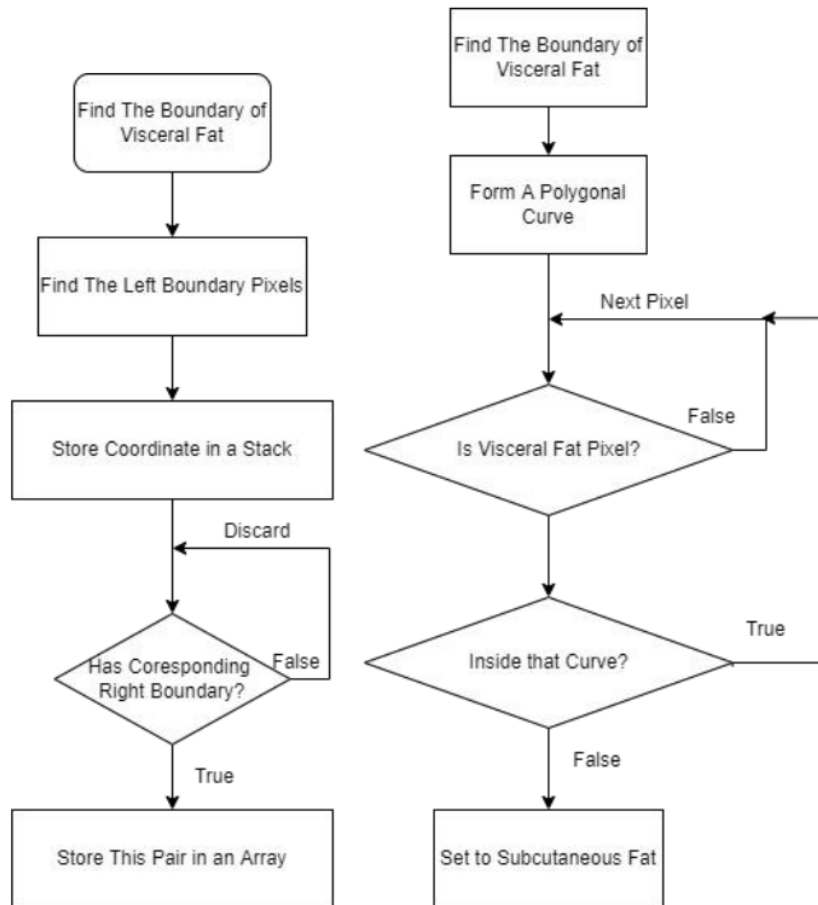


Chart 3.4

There is a very important concept in this sub-algorithm: finding a inside wall pair. During the loops, if a left wall pixel do not have a according right wall pixel, then this left wall pixel is discarded.

Next step is to draw a polygon in order of the vertical coordinate, left part is in order and right part is in a reverse order. The fat tissue inside this polygon are assigned into visceral fat and the rest is subcutaneous fat. The result is shown in [Figure 3.7](#).

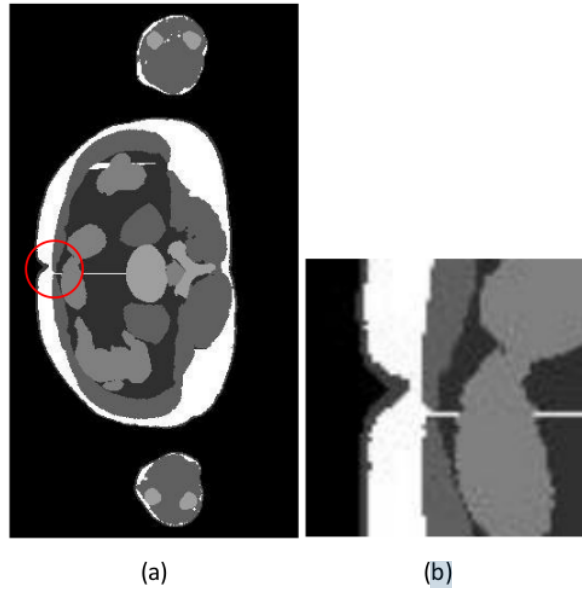


Figure 3.7 Illustration about the Gap Between Muscle Pixels and the Flawed Output

Some side-effect of this method is because of the gap between two part of muscles (the red circle in Figure 3.8). A conditional judgement can be added in initial polygon algorithm. The new polygon algorithm is shown in Chart 3.5.

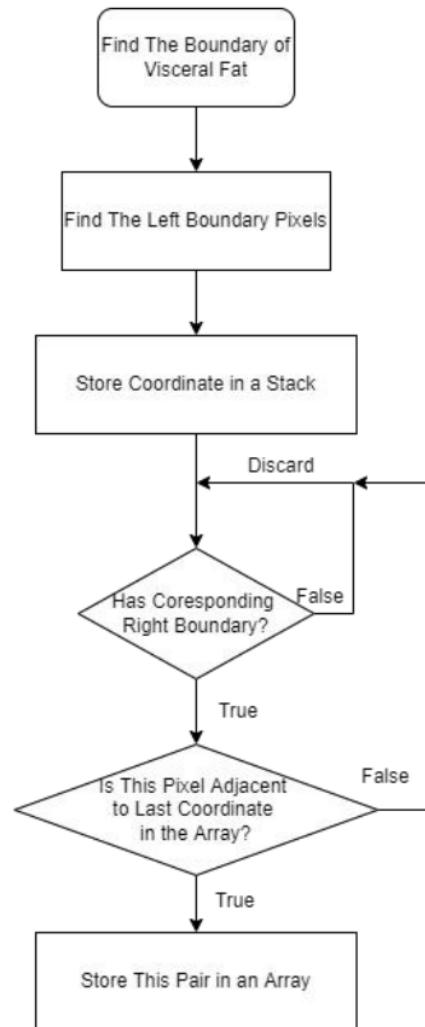


Chart 3.5

Variable *flag* is used to note down the horizontal distance between two left wall pixels. If this distance is bigger than threshold, the second pixel are regarded as another part of muscle. It should not be included in the polygon. By changing threshold, good results can be obtained.

4 Fat Expansion Algorithm (FEA)

4.1 The Method to Calculate the Expanded Value of Fat

For applying the fat expansion algorithm in a more realistic way, the 'quantity' of fat should be found. This will affect the size of chosen filter.

Body mass index (BMI) is a important and major scale to judge whether this person overweight or not. The definition of BMI is the ⁷weight(in kg) divided by the square of the height(in cm). But in this project, the weight and the height is not clear, which means BMI is not useful in this case.

In medical domain, the 'quantity' of fat is also calculated by the volume of fat tissue in liters. Since the resolution of data image is known and the distance between two adjacent slices is known (both 1mm/pixel), the volume of fat expansion is linked with pixels, which is helpful to find a proper filter. Without regarding obese patient, which means it is more unreliable to use MRI to describe[22], a normal overweight person has 2 more liters of visceral fat than normal person approximately[23]. 2 liters are equal to $2 \times 10^3 \text{ mm}^3$, so 2×10^3 pixels should be increased outside the organs for the whole body.

4.2 Fat Expansion Algorithm (FEA)

4.2.1 Methodology of FEA

In this project, fat expansion algorithm (FEA) uses a isotropic method. In another word, FEA in this project utilizes dilation method mentioned in the previous chapter. By increasing the new visceral fat pixels outside original visceral fat, the new image is regarded as expanded image. This process is considered as pressing the other tissue.

There are 4 parts of tissue that connected with visceral fat, which are subcutaneous fat, organs, muscles and bones respectively. Among them, subcutaneous fat make no influence of final result because subcutaneous and visceral fat are similar in substance composition. Visceral fat will only expands in three direction: fat-bones, fat-organs and fat-muscles.

Now the problem converts into how many pixels will visceral fat expanded. Different tissues have different elastic properties[24]. In this project, bones are regarded as stable and all

organs have the same elastic property. Muscles are more easily to be compressed comparing with bones but they are harder to be compressed rather than organs. The model is built as followed:

$$\begin{cases} n_m + n_o = N \\ R * n_m + n_o = E \end{cases}$$

Formula 4.1

where n_m and n_o are the number of pixels on the boundary between visceral fat and muscles or organs, N represents the number of boundary pixels in total, E is the amount of expansion visceral fat and R is the ratio parameter.

By setting up the ratio value, the width of pixels expanded in fat-organ direction and fat-muscle direction can be determined. This width equals to the radius of filter used.

In the end, using dilation method to expand the fat pixels. In this project, the other changes will not be applied, such as the expansion of subcutaneous fat, since they make no contribution to the visceral fat part. The replaced muscle pixels and replaced organ pixels are recorded in two arrays, which means it provides a port with other algorithms.

4.2.2 Finding Edge Pixels of Visceral Fat and Determined the Filter Size

This part will mainly focus on finding the edge of visceral fat. From last section, expansion pixels only occur at the junction between visceral fat and tissues(muscles and organs). Because the resolution is 1mm per pixel, from the whole body perspective, one pixel can be considered as 1mm³. If one pixel's neighbour pixels have one or more organ pixels, this pixel is noted in a fat-organ array. If its neighbour pixels have one or more muscle pixels, this pixel is noted in a fat-muscle array. The results are shown in Figure 4.1.

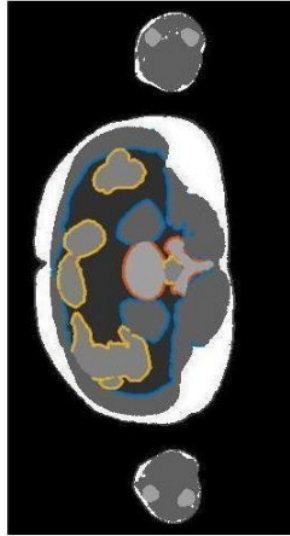


Figure 4.1 The Boundary between Visceral Fat and Other Tissues (Blue Curve is vis-muscle Boundary, Yellow Curve is vis-organ Boundary and Red Curve is vis-bone Boundary)

Apply this method in all the slices and count down the total pixels in fat-organ array and fat-muscle array separately. Then by utilizing Formula 4.1, the size of filter can be found.

4.2.3 Fat Expansion Algorithm

Dilation method is used in this algorithm with structure element mentioned in last section (radius determined filter). New visceral fat pixels substitute the original tissue pixels (organs or muscles). This algorithm is very easy to apply according to the dilation operation mentioned in last chapter.

4.3 Results in Different Direction and GUI Design

4.3.1 Plot Processed Images in Non-transverse Plane

In medical field, there are three planes used to describe the human body: transverse plane, sagittal plane and coronal plane (as shown in Figure 4.2).

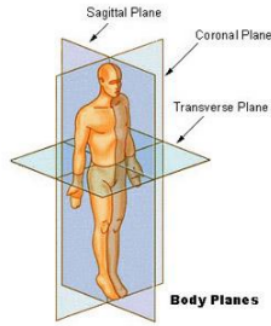


Figure 4.2 Body Planes in Anatomy Field [25]

Typically, MRI scans along the transverse plane. But it is simple to find that sagittal plane and coronal plane are more easily to detect body's shape. By storing all processed slices data and read them into a 3D matrix, different slice of different body plane can be obtained.

4.3.2 GUI Design

In order to make readers and users have a more visualized sensation, a good GUI should be designed.

Users are allowed to change several important variables such as the volume of expanded fat and the elasticity ratio between organs and muscles. After processing all the data, users can select any body plane and any slice of this plane. Also, original slice should be shown as a comparison. A demo of this GUI is shown in Figure 4.3.

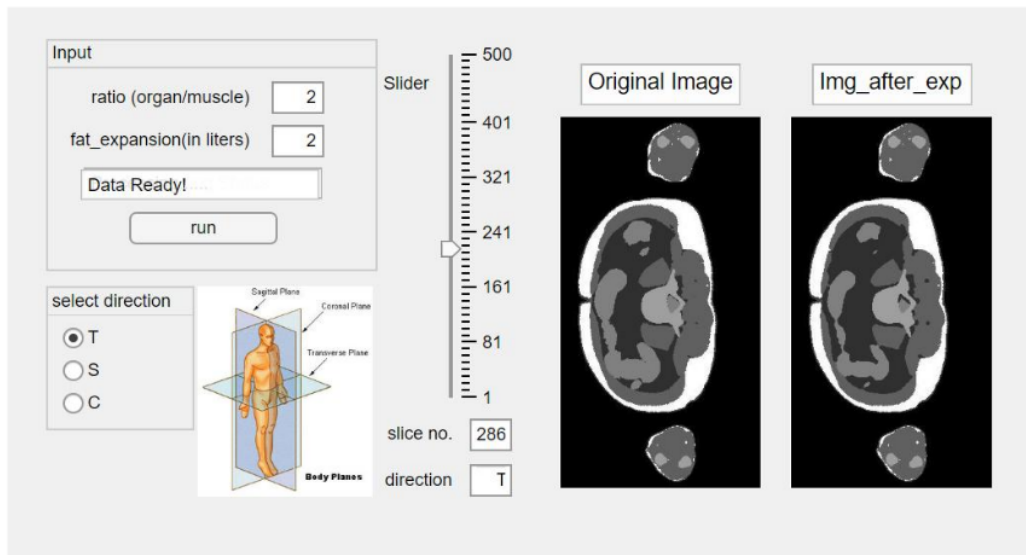


Figure 4.3 The Demo of GUI in This Project

5 Results and discussion

5.1 Results after Reassign the Gray Values

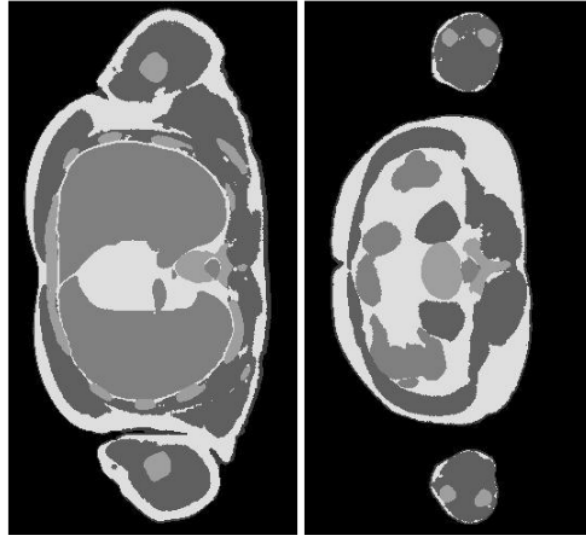


Figure 5.1 Images after Reassigned Values of Tissues (Slice 1 and 300)

Two images in Figure 5.1 are already shown in Figure 3.1. For the convenience for comparison with other outcomes, they are plot again here. Note that the light gray parts means the fat tissue, including visceral fat and subcutaneous fat.

By reassign the gray values of original images, different tissues can be recognized very easily. All the future work is depended on the reassigned images. Now, in the later sections in this chapter, reassigned images will be called original image.

5.2 Results of Fat Distinguish Algorithm

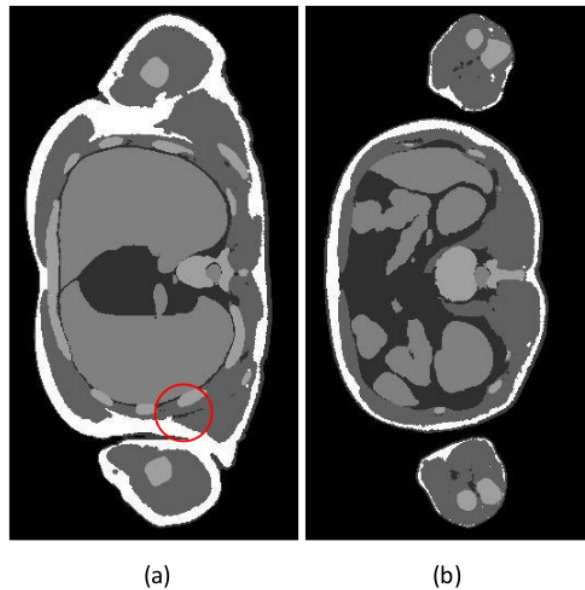


Figure 5.2 Results of FDA in Chest Part (Slice 1 and 200)

From Figure 5.2, ³ it can be seen by comparing two images that for the chest part (from the first slice to 230th slice), fat distinguish algorithm have a very high accuracy rate. Only very small amount pixels could not be detected correctly, as the red circle in Figure 5.2a shown.

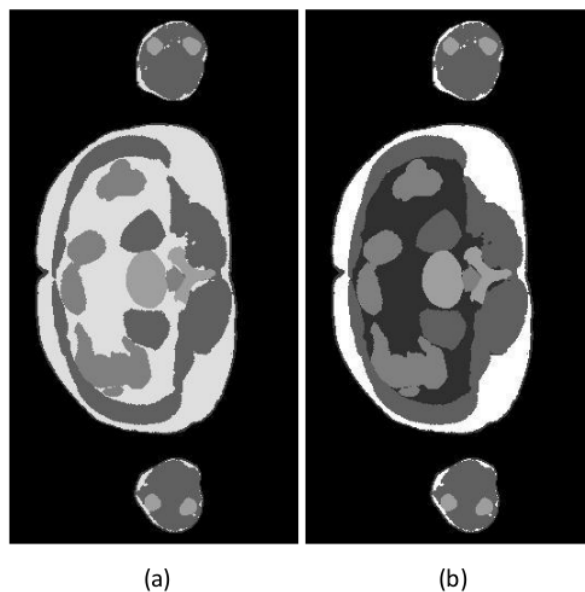


Figure 5.3 Result of FDA at Top Abdomen Part (Slice 300)

Figure 5.3 shows the comparison of original image and distinguished image at top abdomen part. In this slice, only one pixel are unsuccessfully distinguished. But for the bottom abdomen part, the imperfect pixels will increase a little, as Figure 5.4 shown.

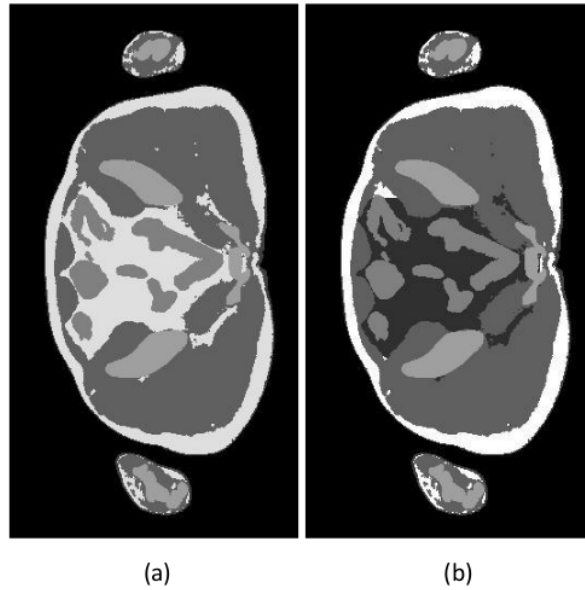


Figure 5.4 Imperfection Results for FDA (Slice 400)

The reason causing this is using the same flag parameter for all the abdomen slices, which cause the imperfection results in polygon improvement algorithm.

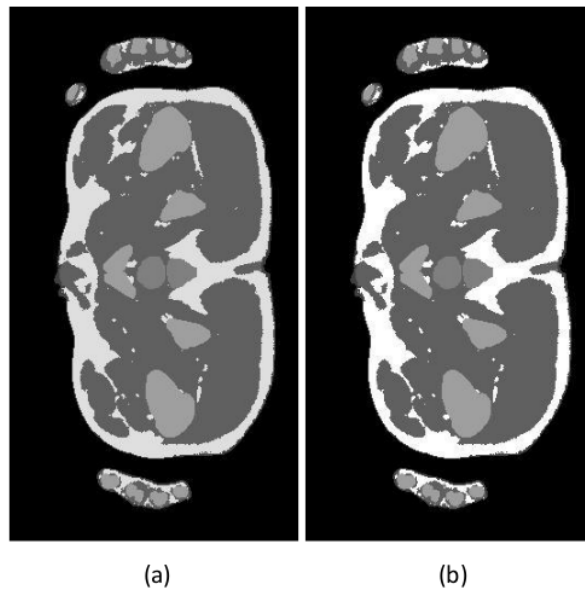


Figure 5.5 Results for Last 30 Slices (Slice 475)

Figure 5.5 shows one example of the last 30 slices, according to the anatomy research, the fat in this part should be subcutaneous fat [18].

Overall, although there are some flaws in this model due to the choice of flag parameter, this fat expansion algorithm can distinguish visceral fat and subcutaneous fat precisely. The fat expansion algorithm is based on this section's work.

5.3 Results of Fat Expansion Algorithm

From last chapter, all the visceral fat and subcutaneous fat are distinguished. The fat expansion algorithm (FDA) is based on the results of fat distinguish algorithm (FEA). So in this section, the comparison is between the results of FDA and the results of FEA. The volume of expanded visceral fat equals to 2 liters and the ratio parameter used in this section is equal to 2 in the illustrations in this section.

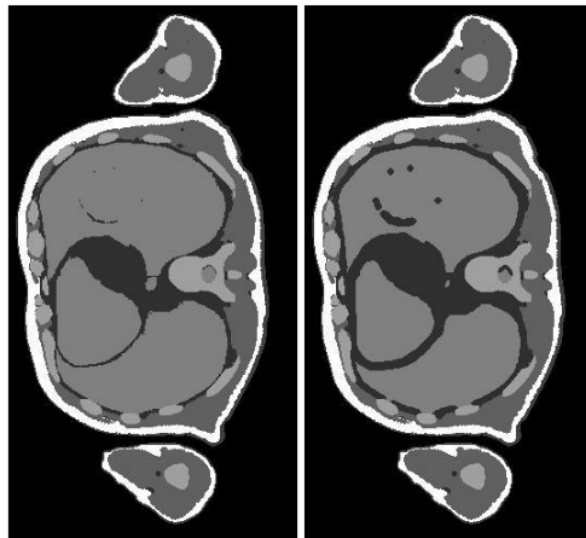


Figure 5.6 Comparison of Results of FDA and FEA in Chest Part (Slice 75)

Figure 5.6 illustrates the image after FEA for the chest part. There are two organs in upper torso, one is lungs, the other is the heart. One characteristic in chest part is that the total amount (pixels) of visceral fat is small, which also causes the number of visceral fat outside wall pixels is small. So the influence of the expansion of visceral fat here is not as significant as the results in abdomen part, as shown in Figure 5.7.

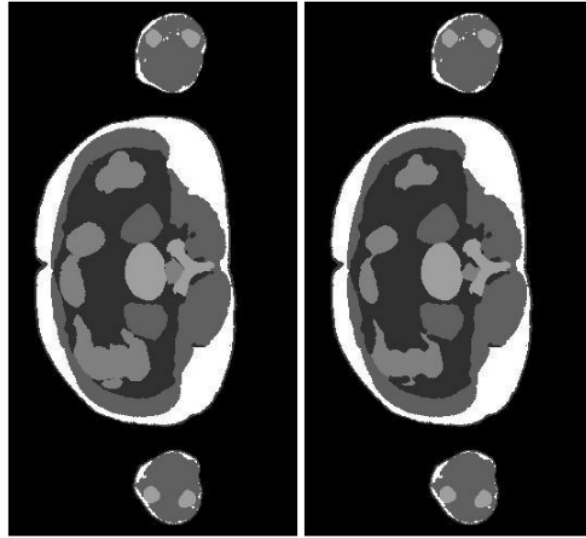


Figure 5.7 Comparison of Results of FDA and FEA in Abdomen Part (Slice 300)

For this part, visceral fat have more junction pixels with muscles and organs. So the influence of FEA is more remarkable than the chest part.

For the last 30 slices, since there are no visceral fat tissue due to the anatomy research, the results in those slices should be exactly as same as the FDA results.

5.4 The Comparison between FDA Algorithms

The figure below shows the performance of initial FDA and two sub-algorithm: label method and polygon method applied in both chest part and abdomen part. Ratio parameter is set to 2 and the expansion volume is set to 2 liters in this section.

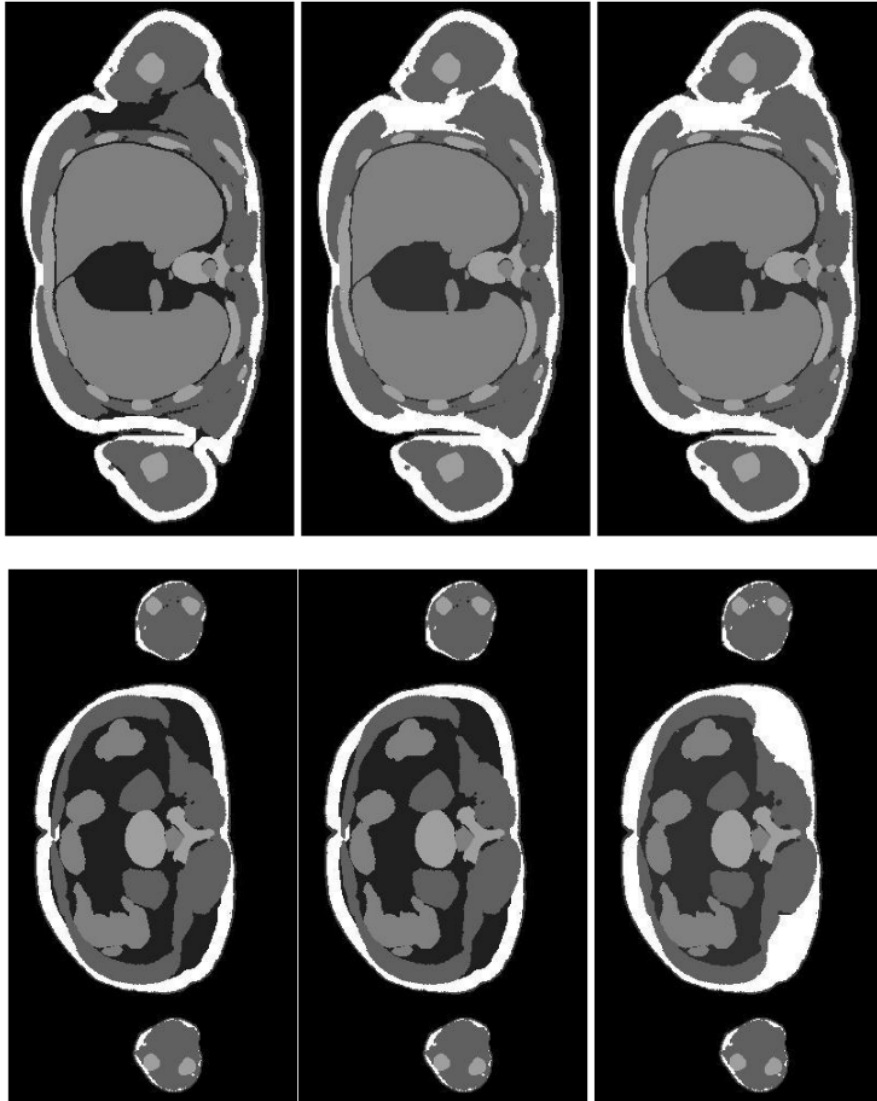


Figure 5.8 The First Row are Slice 1 and The Second Row are Slice 300. The First Column are The Results for Initial FDA and The Last Two Column are The Results for Two Sub-Algorithms

By comparing Figure 5.8b and Figure 5.8e, it is clear that the fat connected with both original subcutaneous fat and original visceral fat can not be distinguished by label method, this always happened in abdomen part. So polygon method should be included in abdomen distinguishing. Although polygon method can also applied in chest part of the body, their calculations are much more than label method. Polygon method requires to find the outside wall of visceral fat and needs to set the flag parameter, while label method only need to apply a label function and compare current the label of current pixel and target label.

Hence, FDA should be divided into three sections according to different parts of the torso, which are chest part, abdomen part and the bottom abdomen part. Actually, in medical field, the chest part and the abdomen part are always regarded separately. Initial FDA is applied in all three parts. Then label method is helpful for chest operation and polygon method is used for improving the performance of initial FDA. For the bottom abdomen part, all the fat are considered as subcutaneous fat.

5.5 The Comparison between Results of FEA in Different Parameters

There are two parameters used in FEA, which are ratio parameter and the volume of expanded visceral fat. Although the ratio parameters vary from different individuals and different organs, they are set to a same value for simple simulation. The quantity of expanded visceral fat determined what is the extent of obesity for a man. In this project, the volume should be set between zero to two liters. The results of different parameters are shown in Figure 5.9.

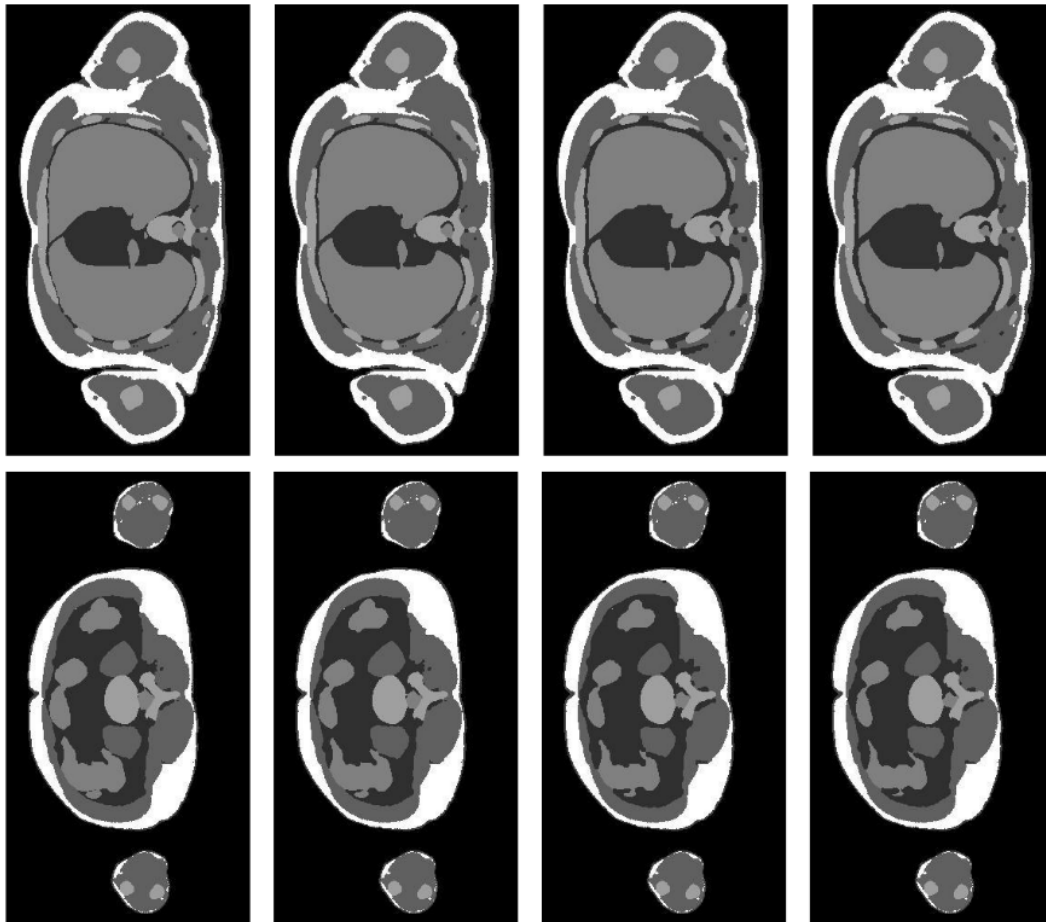


Figure 5.9 The Results under Different Parameters for Slice 1 and Slice 300. Four Column have Parameters Equal to [1,1], [1,2], [2,1] and [2,2] Respectively.

By comparing those outcomes, it is easy to notice that the expansion quantity can infect the overall shape of the body, while the ratio parameters determine the extent of organs or muscles being pressed.

5.6 Results from Other Directions

According to Chapter 4, there are three directions in anatomy, which is sagittal plane, coronal plane and transverse plane. Previous results are all on transverse plane, but it is more visualized to see the results from the other two planes. The results are shown in Figure 5.10 under the parameters are set to 2 and 2.

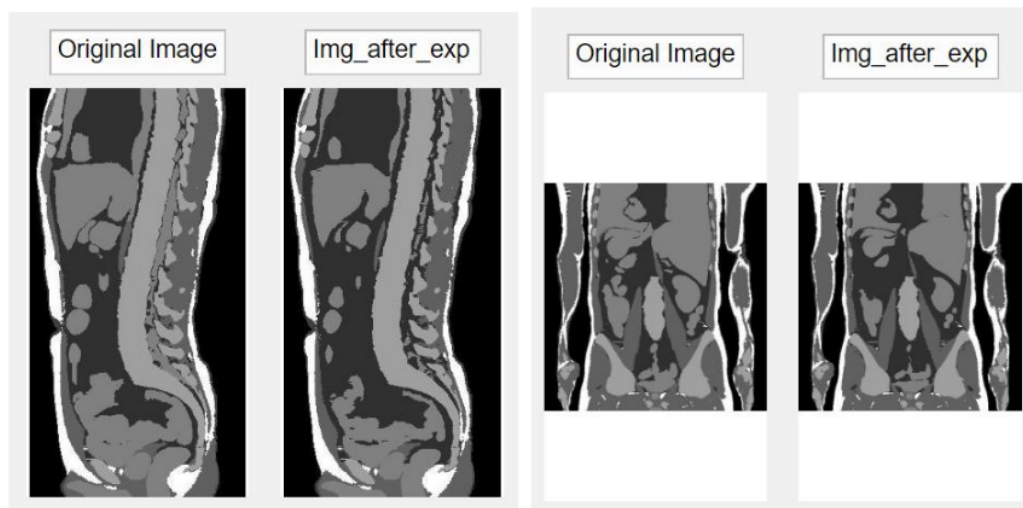


Figure 5.10 The Results of FEA in Other Directions. (Obtained from GUI) (Left Figure is The Result from Slice 240 of Sagittal Plane and The Right One is The Result from Slice 118 of Coronal Plane)

Since the expansion of subcutaneous fat is not considered in this project, the overall shape of the body seems not change at all. But it is totally different for the circumstances inside the subcutaneous fat. From the Figure 5.10 shows, organs are pressed a lot due to the expansion of visceral fat. Some unrealistic output are presented by setting the parameters to 2 and 2 (some tissues are pressed to a non-real shape), because the ratio parameter should not all the same for all the organs and 2 more liters of visceral fat is a very big amount for a overweight person.

Also the imperfections in polygon method is more noteworthy from the view of sagittal plane and coronal plane. This is because of the continuity of visceral fat tissue, which a constant value for flag parameter can not distinguish the visceral fat and subcutaneous fat in several continuous slices.

5.7 The Comparison between Output of FEA and the MRI of a Real Overweight person

The final results of this project is created from a healthy normal man instead of a real MRI from a overweight person. The only way to judge the performance of those results is to compare the results with the real overweight person's MRI by watching the tissues inside the subcutaneous fat. Due to the different characteristics, the chest part and the abdomen part should be checked separately. The comparisons are shown in Figure 5.11.

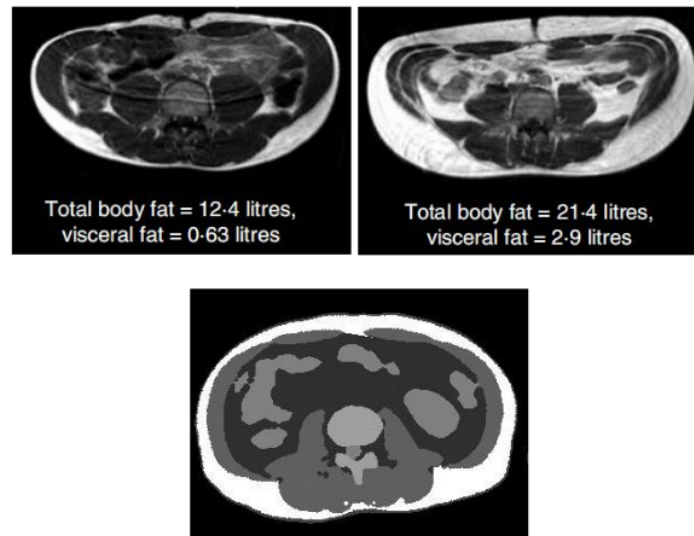


Figure 5.11 Comparison between the MRI for Real Overweight Patient [23] and the Outcome after FEA. (The arm part of Outcome of FEA is Ignored and This Figure are Rotated for Better Comparison)

From the comparison figures shows in Figure 5.11, by seeing the first row, when the visceral fat increasing, the organs and the muscles are pressed. The outcomes simulates this feature very well. But in FEA, fat is expanded in a circular manner, while in real situation, fat always expands in a non-uniform manner. This imperfection should be concerned in the future work.

6 Future Improvements and Future Work

6.1 Future Improvements

6.1.1 Set Specified Flag Parameters for Different Slices

From last chapter, the imperfection of FDA is shown, which is caused by using the same value for flag parameter in every slices. In order to solve this problem, the specified flag parameters should be applied in the polygon method. Or some other method should be introduced to improve initial fat distinguish algorithm.

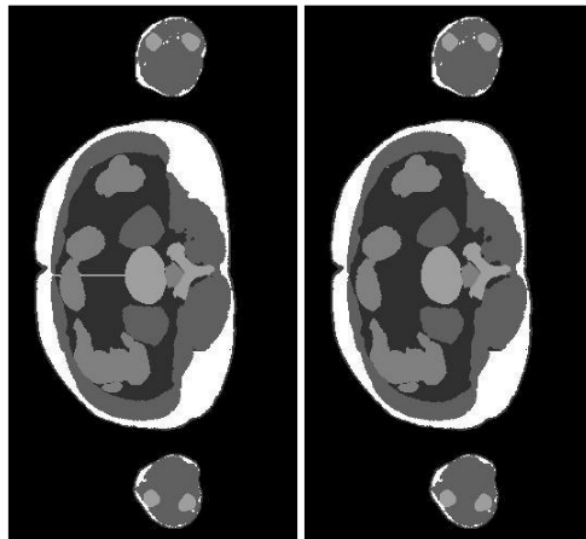


Figure 6.1 The Results of FDA when Flag Parameter Equals to 130 and 140

See from Figure 6.1, even slight change of flag parameter could infect the result a lot. So the optimal flag parameter for every slice should be found artificially slice by slice and then put those values in a array.

6.1.2 Use Euclidean Distance instead of Disk Shape Filter

In this project, the structure element used is generated by a existing function. But when the radius of this disk filter increases, the filter becomes more likely a octagon shape, which will cause the abnormal output [9]. Instead, circular structure element generated by calculating the euclidean distance should be used.

By using euclidean distance to generate circular filter, the real circular expansion can be simulate more precisely when the radius of the filter is large (to simulate the obesity patient).

6.1.3 Set the Expansion Volume Separately for Chest Part and Abdomen Part

The quantity of expanded visceral fat is considered all over the body. However, in real medical field, the chest part and the abdomen part show the different potential of fat expansion. Typically, the abdomen part are more likely to accumulate the visceral fat rather than the chest part. Also the total amount of visceral fat in the abdomen part is larger than the chest part.

6.1.4 Set Specified Ratio Parameters for Different Organs

After dividing expansion quantity in two parts of the body, the expansion tendency for different organs also need specified values. In the future project, those values could be a set of ratios compared with expansion tendency for muscles. For example, if one organ has the same elasticity with the muscles surround it, the ratio parameter for this organ should be set to one. By do so, a more realistic output can be generated. Both this section and the previous section need a lot of medical research.

Then, since the original data is collected from a healthy 55 years old Asian man, different sets of ratio parameters can be used for generating various target output. The elasticity can obtained by the study of anatomy on different ages, different races and different genders.

6.1.5 Change the Muscle Pixels

In this project, the position changes of muscle pixels are mostly ignored, instead, the muscle pixels connected with visceral fat are replaced directly by visceral fat pixels. However, the total amount of muscle should not change. So in the future project, the muscle pixels changing should be obtained along with subcutaneous fat expansion in order to get a complete fat expansion image.

6.2 Future Work

6.2.1 Apply Subcutaneous Fat Expansion afterwards

One more important thing to generate fat expansion images is to add the subcutaneous fat expansion works after the visceral fat expansion. Subcutaneous fat expansion should operate later than visceral fat expansion because in visceral fat expansion part, the positions of muscle pixels are changed, which will infect a lot on the final output of subcutaneous fat expansion.

6.2.2 Nonuniform Expansion

A uniformed expansion are applied in this project. But for a real human body, a circular expansion is not realistic. Some fat tissues are highly depend on the tissue and cells surround them. This part of improvement will include more medical research and more advanced image processing method such as the object detection under some deep learning model.

6.2.3 Obesity instead of Overweight

According to chapter 4, only 2 liters in maximum of visceral fat are considered. But a real obesity patient obtains much more volume of visceral fat than overweight person which is considered in this project. In the future work, more quantity of visceral fat could be added.

This part future work will cause some problems when using the same model in this project. For example, if the radius of the circular filter is too big, some tissues will appear a abnormal form. For instance, all the pixels connected with some bones are visceral fat pixels, which is unable to happen in a real person. So a new method or other improve sub-algorithm should be developed when the obesity patient is the target.

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Appendices

A Code

The programs for this project are available via: <https://github.com/Cn-Gty/Fat-Expansion>

B Project Outline

The outline of this report is listed as followed:

- 1) Chapter 2 is the literature survey. This chapter includes all the necessary knowledge related to this project such as the format of medical image and the anatomy research.
- 2) Chapter 3 is the first section of this project: Fat Distinguish Algorithm (FDA). Initial algorithm and two sub-algorithms are put forward to distinguish subcutaneous fat and visceral fat.
- 3) Chapter 4 is the second section of this project: Fat Expansion Algorithm (FEA). FEA is introduced in this chapter and the GUI is also mentioned in the last section of this chapter.
- 4) Chapter 5 is the results and discussion. This chapter shows all the outcomes (intermediate results and final results) from the program. The imperfections in the project are evaluated and the discussions around those imperfections are made.
- 5) Chapter 6 is the future work. The solutions for the imperfections of algorithms are put forward. In the end, the striving direction of future works are pointed out.

C Risk assessment

Risk assessment is at next two pages.

1 SCHOOL OF E&EE RISK ASSESSMENT

1 WORK ACTIVITY/ WORKPLACE (WHAT PART OF THE ACTIVITY POSES RISK OF INJURY OR ILLNESS)	HAZARD (S) (SOMETHING THAT COULD CAUSE HARM, ILLNESS OR INJURY)	LIKELY CONSEQUENCES (WHAT WOULD BE THE RESULT OF THE HAZARD)	WHO OR WHAT IS AT RISK (INCLUDE NUMBERS AND GROUPS)	EXISTING CONTROL MEASURES IN USE (WHAT PROTECTS PEOPLE FROM THESE HAZARDS)	WITH EXISTING CONTROLS			WITH NEW CONTROLS		
					SEVERITY	LIKELIHOOD	RISK RATING	RISK ACCEPTABLE	SEVERITY	LIKELIHOOD
Spend prolonged periods sitting before computer/laptop and dragging a mouse	Repetitive Strain Injury (RSI) and Work Related Upper Limb Disorder (WRULD)	Pain, Numbness, Aching, Tiredness and Tingling in muscles, nerves, tendons and other soft body tissues of the hand, wrist, arm, shoulder and spine	Students who need to program and use computer and laptop in the project	Students are to be provided with copies of the documents of Health and Safety Executive publication "INDG36 - Working with VDUs".	2	3	6	N O	2	1
Focusing the eyes on a computer display for protracted, uninterrupted periods of time	Computer vision syndrome (CVS)	Headaches, Blurred vision, Neck pain, Fatigue, eye strain, Dry, Irritated eyes, Double vision, Polyopia, and Difficulty refocusing the eyes	Students who need to program and use computer and laptop in the project	Approved seating, desks and lighting are provided in laboratory SSB/A 19 Students are to be provided with copies of the following documents: (1) The University's Code of Practice and Guidance Note on "Use of Display Screen Equipment", (2) The University's Guidance Note "Display Screen Equipment / Workstation Set Up"	3	3	9	N O	2	1

Activity Location: Home

MANAGER/SUPERVISOR	NAME: Dr. Costen	SIGNED:	DATE:
STUDENT	NAME: Taiyue Guo	SIGNED: Taiyue Guo	DATE: 04/09/2022

1

SCHOOL OF E&EE RISK ASSESSMENT

IF THE ANSWERS TO ANY OF THE QUESTIONS BELOW IS YES THEN ADDITIONAL SPECIFIC RISK ASSESSMENTS MAY BE REQUIRED:

IS THERE A RISK OF FIRE?	Y/N	DOES THE ACTIVITY REQUIRE ANY HOME WORKING?	Y/N
ARE SUBSTANCES THAT ARE HAZARDOUS TO HEALTH USED?	Y/N	ARE THE EMPLOYEES REQUIRED TO WORK ALONE	Y/N
IS THERE MANUAL HANDLING INVOLVED?	Y/N	DOES THE ACTIVITY INVOLVE DRIVING	Y/N
IS PPE WORN OR REQUIRED TO BE WORN?	Y/N	DOES THE ACTIVITY REQUIRE WORK AT HEIGHT	Y/N
ARE DISPLAY SCREENS USED?	Y/N	DOES THE ACTIVITY INVOLVE FOREIGN TRAVEL	Y/N
IS THERE A SIGNIFICANT RISK TO YOUNG PERSONS?	Y/N	IS THERE A SIGNIFICANT RISK TO NEW / PREGNANT MOTHERS?	Y/N

Severity value = potential consequence of an incident / injury

5	Very High	Death / permanent incapacity / widespread loss
4	High	Major Injury (Reportable Category) / Severe Incapacity / Serious Loss
3	Moderate	Injury / illness of 3 days or more absence (reportable category) / Moderate loss
2	Slight	Minor injury / illness – immediate First Aid only / slight loss
1	Negligible	No injury or trivial injury / illness / loss

Likelihood value = what is the potential of an incident or injury occurring

5	Almost certain to occur
4	Likely to occur
3	Quite possible to occur
2	Possible in current situation
1	Not likely to occur

Risk rating = severity value x likelihood value

Risk ratings are classified as low (1 – 5), medium (6 – 9) and high (10 – 25)

Risk Classification and Actions:

Rating	Classification	Action
1 – 5	Low	Tolerable risk - Monitor and Manage
6 – 9	Medium	Review and introduce additional controls to mitigate to "As Low As Reasonably Practicable" (ALARP)
10 – 25	High	Stop work immediately and introduce further control measures

SEVERITY

	1	2	3	4	5
1	Low	Low	Low	Low	Low
2	Low	Low	Medium	Medium	High
3	Low	Medium	Medium	High	High
4	Low	Medium	High	High	High
5	Low	High	High	High	High

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